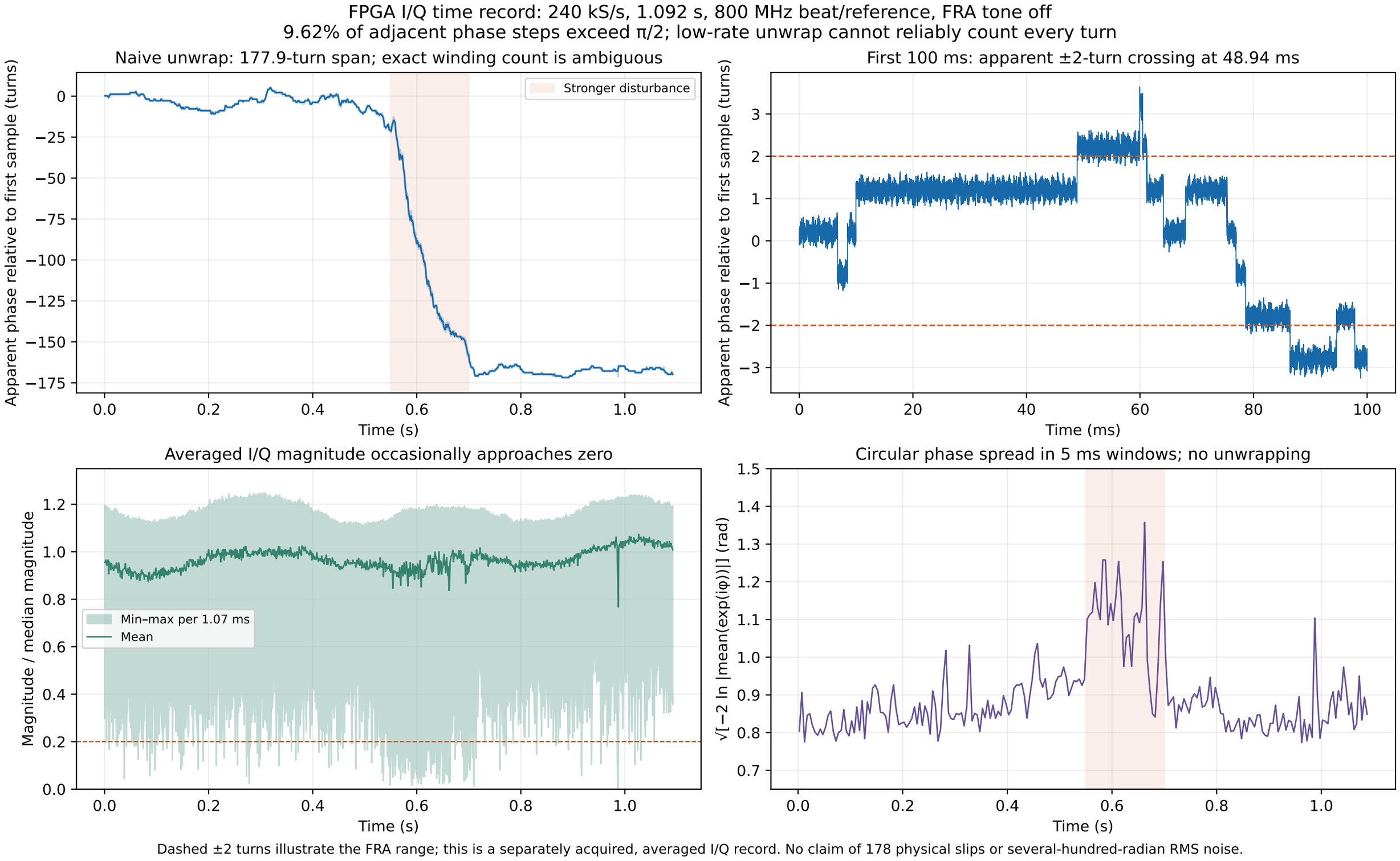

FPGA I/Q time record: 240 kS/s, 1.092 s, 800 MHz beat/reference, FRA tone off
9.62% of adjacent phase steps exceed π/2; low-rate unwrap cannot reliably count every turn
Naive unwrap: 177.9-turn span; exact winding count is ambiguous
Stronger disturbance
Apparent phase relative to first sample (turns)
Time (s)
First 100 ms: apparent ±2-turn crossing at 48.94 ms
Apparent phase relative to first sample (turns)
Time (ms)
Averaged I/Q magnitude occasionally approaches zero
Min–max per 1.07 ms
Mean
Magnitude / median magnitude
Time (s)
Circular phase spread in 5 ms windows; no unwrapping
√[−2 ln |mean(exp(iφ))|] (rad)
Time (s)
Dashed ±2 turns illustrate the FRA range; this is a separately acquired, averaged I/Q record. No claim of 178 physical slips or several-hundred-radian RMS noise.